

citemsp Package

Per-Key Citation Locators

v2.3 — July 23, 2026

Apostolos Tsampodimos  ¹, Nathaniel Sherrill², and Agnese Mariotti²

¹Center for Physical Sciences and Technology, Saulėtekio 3, 10257 Vilnius, Lithuania,
apostolos.tsampodimos@ftmc.lt

²Institute for Theoretical Physics, Leibniz Universität Hannover, Appelstraße 2, 30167
Hannover, Germany

Abstract

The `citemsp` package provides a single user-facing command, `\citemsp`, that attaches section (§), paragraph (¶), and prefix-based locators directly to numeric citation labels. It works on top of either `biblatex` or `natbib`, auto-detecting which backend is loaded. An extensible prefix registry provides, currently, ten built-in types and users can register custom prefixes with `\citemspprefix`. The package parses a slash-delimited syntax, rendering each locator as a superscript/subscript pair on the corresponding citation number, producing output such as $[9_{45}^{§3.2}, 11^{pp.520}, 7_{eq.4.3}^{§1}]$. Plain entries (no slashes) are batched and passed to the backend's native `\cite`, so range compression (`numeric-comp`, `sort&compress`) works naturally.

1 Motivation

The `biblatex` package offers a locator interface through its postnote field. A user can write `\autocite[3.2]{smith2003}` to append a free-form locator after the citation number. For multi-key citations, `\autocites` accepts per-key pre- and post-notes:

```
\autocites[3.2]{smith2003}[¶2]{meyerrochow2003}
```

This approach has two limitations when writing a scientific paper:

1. **Verbose syntax.** Citing three sources with locators requires nested optional arguments, which is cumbersome in source files.
2. **No typed locators.** There is no compact notation for referencing specific equations, figures, definitions, or theorems within cited works.

The `citemsp` package renders locators as a superscript/subscript pair attached to the citation number, supports ten built-in prefix types via an extensible registry, and accepts all keys with their locators in a single comma-separated argument. It supports both `biblatex` and `natbib` backends.

2 Installation

2.1 Per-project

Copy `citemsp.sty` into the same directory as your main `.tex` file.

2.2 System-wide (local texmf)

```
mkdir -p ~/texmf/tex/latex/citemsp
cp citemsp.sty ~/texmf/tex/latex/citemsp/
texhash ~/texmf
```

3 Usage

3.1 Loading with biblatex

The package must be loaded *after* `biblatex`:

```
\usepackage[style=numeric, backend=biber]{biblatex}
\addbibresource{refs.bib}
\usepackage{citemsp}
```

3.2 Loading with natbib

```
\usepackage[numbers]{natbib} % omit if your class loads it
\usepackage{citemsp}
```

Classes that auto-load `natbib` (e.g., `revtex4-2`, `aastex631`) require no explicit `\usepackage{natbib}`. Loading `citemsp` before either backend produces a package error.

3.3 Syntax

The `\citemsp` command accepts a comma-separated list of entries. Each entry has the form `key`, `key/loc1`, or `key/loc1/loc2`. By default, the first slot renders as § (section) and the second as ¶ (paragraph). A single-letter prefix changes the locator type:

<code>\citemsp{key}</code>	→	$[9]$
<code>\citemsp{key/section}</code>	→	$[9^{\S 6.2}]$
<code>\citemsp{key/section/paragraph}</code>	→	$[9^{\S 6.2}_{\P 3}]$
<code>\citemsp{key//paragraph}</code>	→	$[9_{\P 3}]$
<code>\citemsp{key/section/e4.3}</code>	→	$[9^{\S 6.2}_{\text{eq. 4.3}}]$

Note the subscript-only form `key//par`: an empty first slot with a double slash gives a paragraph (or prefixed) subscript without a section superscript.

3.4 Prefix registry

All ten built-in prefixes are listed below. Note that no prefix defaults to § (section) in the superscript slot and ¶ (paragraph) in the subscript slot. A prefix is a single letter at the start of a locator slot that selects a symbol:

Prefix	Locator	Render	In super slot	In sub slot	Both slots
A	Appendix	App.	$[9^{\text{App. 2}}]$	$[9_{\text{App. 2}}]$	$[9^{\S 3.2}_{\text{App. 2}}]$
C	Corollary	Cor.	$[9^{\text{Cor. 3}}]$	$[9_{\text{Cor. 3}}]$	$[9^{\S 4.1}_{\text{Cor. 3}}]$
L	Lemma	Lem.	$[9^{\text{Lem. 1}}]$	$[9_{\text{Lem. 1}}]$	$[9^{\S 3.2}_{\text{Lem. 1}}]$
T	Theorem	Th.	$[9^{\text{Th. 4.2}}]$	$[9_{\text{Th. 4.2}}]$	$[9^{\S 6.1}_{\text{Th. 4.2}}]$
d	Definition	$\triangleq$	$[9^{\triangle 5}]$	$[9_{\triangle 5}]$	$[9^{\S 3.2}_{\triangle 5}]$
e	Equation	eq.	$[9^{\text{eq. 4.3}}]$	$[9_{\text{eq. 4.3}}]$	$[9^{\S 6.2}_{\text{eq. 4.3}}]$
f	Figure	fig.	$[9^{\text{fig. 3}}]$	$[9_{\text{fig. 3}}]$	$[9^{\S 6.1}_{\text{fig. 3}}]$
n	Footnote	†	$[7^{\dagger 7}]$	$[7_{\dagger 7}]$	$[7^{\S 1}_{\dagger 7}]$
p	Page	pp.	$[9^{\text{pp. 42}}]$	$[9_{\text{pp. 42}}]$	$[9^{\S 3.2}_{\text{pp. 42}}]$
t	Table	$\boxplus$	$[11^{\boxplus 2}]$	$[11_{\boxplus 2}]$	$[11^{\S 8.4}_{\boxplus 2}]$

Table 1: Built-in prefix registry.

Users can register custom prefixes with `\citemspprefix`. Place the command in the preamble (after `\usepackage{citemsp}`) so it is available throughout the document:

```
\usepackage{citemsp}
\citemspprefix{R}{Rem.}      "R" renders as "Rem."
\citemspprefix{w}{w.}       "w" renders as "w."
```

For example, after registering R for “Remark”: `\citemsp{cox1979/12.4/R3}` → $[2^{\S 12.4}_{\text{Rem. 3}}]$, and after registering w for “widget”: `\citemsp{zongker2006/3/w5}` → $[13^{\S 3}_{\text{w 5}}]$. The prefix letter must not collide with an existing built-in prefix.

3.5 Configuration

Two user-configurable parameters control the locator appearance:

- `\citemspscale` (default 0.35) — controls the locator glyph size via `\scalebox` from `graphicx`.

- `\citenspraiseoffset` (default 1.5pt) — vertical offset applied via `\raisebox` to superscript locators.

Redefine them after loading:

```
\renewcommand{\citenspscale}{0.4}
\renewcommand{\citenspraiseoffset}{2pt}
```

4 Comparison with Standard Biblatex

Standard biblatex	citensp
<code>\autocite[§3.2]{smith2003}</code> → [1, §3.2]	<code>\citensp{smith2003/3.2}</code> → [9 ^{§3.2}]
<code>\autocites[§3.2]{smith2003}</code> <code>[eq.~4.3]{meyerrochow2003}</code> → [1, §3.2][2, eq. 4.3]	<code>\citensp{smith2003/3.2,</code> <code>meyerrochow2003/e4.3}</code> → [9 ^{§3.2} , 7 ^{eq.4.3}]
<code>\autocites[§3.2, ¶5]{smith2003}</code> → [1, §3.2, ¶5]	<code>\citensp{smith2003/3.2/5}</code> → [9 ^{§3.2} _{¶5}]
No built-in typed locator syntax; user must type symbols manually	Ten built-in prefixes; extensible via <code>\citenspprefix</code>
Range compression via <code>numeric-comp</code>	Plain entries (no /) compressed via the backend's native <code>\cite</code> , so compression works identically

Table 2: Feature comparison between standard biblatex locators and citensp.

5 Examples

5.1 Single source with section and paragraph

The lack of randomised controlled trials on parachutes~\citensp{smith2003/6.1/2} remains a gap in evidence-based medicine.

Output: The lack of randomised controlled trials on parachutes [9^{§6.1}_{¶2}] remains a gap in evidence-based medicine.

5.2 Section-only locator

Chickens were found to prefer beautiful humans~\citensp{ghirlanda2002/2.3}, raising questions about peer review in poultry science.

Output: Chickens were found to prefer beautiful humans [3^{§2.3}], raising questions about peer review in poultry science.

5.3 Subscript-only locator

An empty first slot (double slash) gives a subscript without a superscript:

```
\citemsp{smith2003//5}           % paragraph 5, no section
\citemsp{meyerrochow2003//e2.1}  % equation 2.1, no section
```

Output: [9_{¶5}], [7_{eq.2.1}].

5.4 Multiple sources with mixed locators

```
Writer's block~\citemsp{upper1974, smith2003/4,
  meyerrochow2003/1/2} has been studied from behavioural,
medical, and ornithological perspectives.
```

Output: Writer's block [10, 9^{§4}, 7_{¶2}^{§1}] has been studied from behavioural, medical, and ornithological perspectives.

5.5 Range compression

Plain entries (no slashes) are batched and passed to the backend's native `\cite`, so range compression works automatically when using `numeric-comp` (biblatex) or `sort&compress` (natbib). Entries *with* locators are rendered individually and interleaved with the compressed group. The examples below illustrate different combinations.

1. All plain (fully compressed):

```
\citemsp{upper1974, pennycook2015, smith2003, meyerrochow2003,
  wang2020}
```

Output: [10, 8, 9, 7, 11]

2. Plain entries mixed with locator entries:

```
\citemsp{upper1974, pennycook2015, smith2003/4.1,
  meyerrochow2003, wang2020}
```

Output: [10, 8, 9^{§4.1}, 7, 11]

The plain entries compress into a range while the locator entry renders separately.

3. Multiple locator entries among plain ones:

```
\citemsp{upper1974, pennycook2015/2.3, smith2003,
  meyerrochow2003/e14, wang2020}
```

Output: [10, 8^{§2.3}, 9, 7_{¶14}^{eq.14}, 11]

4. Every entry has a locator (no compression):

```
\citemsp{upper1974/1, smith2003/4.1/2,
  meyerrochow2003/e14, wang2020/p21}
```

Output: [10^{§1}, 9_{¶2}^{§4.1}, 7_{¶14}^{eq.14}, 11_{pp.21}]

When every entry carries a locator, each is rendered individually and no range compression occurs.

5. Locator entries with super and subscript among plain ones:

```
\citemsp{upper1974, pennycook2015, smith2003/6.1/2,
meyerrochow2003, wang2020, cox1979/12.4/T5}
```

Output: [10, 8, 9^{§6.1}₂, 7, 11, 2^{§12.4}_{Th.5}]

5.6 Equation and figure locators

```
The sloshing dynamics of coffee~\citemsp{mayer2012/3.2/e3.31}
and the critical spill diagram~\citemsp{mayer2012/4.6/f4.1}
explain why your laptop is always sticky.
```

Output: The sloshing dynamics of coffee [6^{§3.2}_{eq.3.31}] and the critical spill diagram [6^{§4.6}_{fig.4.1}] explain why your laptop is always sticky.

5.7 Theorem, lemma, and corollary locators

```
The Cox--Zucker machine~\citemsp{cox1979/12.4/T5} computes
intersection numbers; see Lemma~\citemsp{cox1979/12.3/L2}
and Corollary~\citemsp{cox1979/12.4/C3}.
```

Output: The Cox–Zucker machine [2^{§12.4}_{Th.5}] computes intersection numbers; see Lemma [2^{§12.3}_{Lem.2}] and Corollary [2^{§12.4}_{Cor.3}].

5.8 Definition and page locators

```
The definition of a necrobotic
actuator~\citemsp{yap2022/2.3/d2.1} involves reanimating
dead spiders; see also~\citemsp{yap2022/p37}.
```

Output: The definition of a necrobotic actuator [12^{§2.3}_{§2.1}] involves reanimating dead spiders; see also [12^{pp.37}].

5.9 Appendix and footnote locators

```
The theorem that biologists cannot fix a
radio~\citemsp{lazebnik2002/T11} is proven in
Appendix~\citemsp{lazebnik2002/A2}; the apology to
electrical engineers is in
footnote~\citemsp{lazebnik2002/3/n4}.
```

Output: The theorem that biologists cannot fix a radio [4^{Th.11}] is proven in Appendix [4^{App.2}]; the apology to electrical engineers is in footnote [4^{§3}_{Footnote}].

5.10 Table locator and mixed types in one call

See~\citemsp{smith2003/6.1/2, matthews1995/e6.55,
wang2020/8.4/t2, yap2022/p170} for parachutes, toast,
graphene, and dead spiders respectively.

Output: See [9^{§6.1}_{¶2}, 5^{eq.6.55}, 11^{§8.4}_{⊕2}, 12^{pp.170}] for parachutes, toast, graphene, and dead spiders respectively.

5.11 Plain citation as drop-in replacement

Without slashes, \citemsp behaves identically to \cite:

Elephant dung turns out to be useful
habitat~\citemsp{camposarceiz2009}, which is more than can be
said for most conference proceedings.

Output: Elephant dung turns out to be useful habitat [1], which is more than can be said for most conference proceedings.

5.12 Adjacent to inline math

The penguin defaecation pressure $P \approx 60\,$
\mathrm{kPa}\$~\citemsp{meyerrochow2003/e14, smith2003/4.3}
exceeds most expectations.

Output: The penguin defaecation pressure $P \approx 60$ kPa [7^{eq.14}, 9^{§4.3}] exceeds most expectations.

5.13 Font contexts

The package works correctly in all standard L^AT_EX font contexts. For italic and slanted shapes, a small kern is automatically inserted to prevent the locator from colliding with the slanted closing bracket. Bold, bold-italic, small caps, and upright text work without modification.

Upright: [9^{§6.1}_{¶2}]
Italic: [9^{§6.1}_{¶2}]
Slanted: [9^{§6.1}_{¶2}]
Bold: [9^{§6.1}_{¶2}]
Bold italic: [9^{§6.1}_{¶2}]
SMALL CAPS: [9^{§6.1}_{¶2}]
Emph (italic): [9^{§6.1}_{¶2}] — nested emph (upright): [9^{§6.1}_{¶2}]

5.14 Font sizes

Locator glyphs scale proportionally at different font sizes:

Small: [3^{§2.3}_{eq.2.5}]
Normal: [3^{§2.3}_{eq.2.5}]
Large: [3^{§2.3}_{eq.2.5}]

5.15 Custom prefixes

Place `\citenspprefix` in the preamble, after loading `citemsp`:

```
% in the preamble:
\citenspprefix{w}{w.}      % "w" renders as "w."
\citenspprefix{H}{Hyp.}    % "H" renders as "Hyp."

% in the document body:
\citemsp{zongker2006/3/w5}
\citemsp{cox1979/12.4/H2}
```

Output: $[13_{w,5}^{§§}]$, $[2_{Hyp,2}^{§12.4}]$

6 Natbib Backend

Since v2.2, `citemsp` auto-detects whether `biblatex` or `natbib` is loaded and selects the matching code path. No package option is needed.

6.1 Compatible classes

Many journal classes auto-load `natbib`. With these, simply loading `citemsp` after the class is sufficient:

Class	Backend	Notes
article, report, book, KOMA	either	user's choice
revtex4-2 (APS)	natbib (auto)	class loads natbib
aastex631 (AAS)	natbib (auto)	class loads natbib
elsarticle (Elsevier)	natbib (manual)	add <code>\usepackage[numbers]{natbib}</code>
IEEEtran natbib mode	natbib (manual)	add <code>\usepackage[numbers]{natbib}</code>

Table 3: Document class compatibility.

6.2 Minimal natbib example

```
\documentclass{article}
\usepackage[numbers, sort&compress]{natbib}
\usepackage{citemsp}
\begin{document}
See~\citemsp{key1/3.2/5, key2/e4.1} for details.
\bibliographystyle{unsrt}
\bibliography{refs}
\end{document}
```

On the `natbib` path, `citemsp` uses `\citenum` for the bare number and wraps it in a `\hyperlink` when `hyperref` is loaded. Range compression via `sort&compress` works for plain entries.

7 Implementation

1. **Backend detection.** At load time, the package checks whether `biblatex` or `natbib` is already loaded. If neither is present, a descriptive error is raised. If both are loaded (itself

an error), then `biblatex` is preferred.

2. **Locator scaling.** `\citemsscale` (default 0.35) controls the size of locator glyphs via `\scalebox` from `graphicx`. `\citemspraiseoffset` (default 1.5pt) sets the vertical offset of superscript locators via `\raisebox`.
3. **Font shape detection.** The renderer checks `\f@shape` against `it` (italic) and `sl` (slanted). In either case, a small math kern is inserted to prevent the locator from colliding with the slanted final bracket.
4. **Prefix dispatch.** Each locator value is tested: if its first character matches a registered prefix (via `\csname @citemsp@pfx@... lookup`), the prefix symbol is rendered; otherwise the default § or ¶ is used.
5. **Citation rendering.** On the `biblatex` path, a custom cite command `\citelinksp` (via `\DeclareCiteCommand`) prints the hyperlinked number and attaches locators. On the `natbib` path, `\citenum` provides the bare number, wrapped in `\hyperlink` when `hyperref` is loaded.
6. **Entry parsing.** The user-facing `\citemsp` uses `\forcsvlist` (from `etoolbox`) to iterate over comma-separated entries. Each entry is split at forward slashes using a delimited `\def` pattern, extracting the key and up to two locator slots.
7. **Plain-key batching.** Entries without slashes are accumulated into a buffer and flushed via the backend's native `\cite` (with brackets suppressed), so that range compression (`numeric-comp` or `sort&compress`) works identically to a regular `\cite` call.

8 Limitations

- **Numeric styles only.** `citemsp` attaches locators to numeric citation labels. Other non-numeric styles are currently not supported.
- **Manual paragraph counting.** Paragraph locators refer to paragraph numbers counted manually from the start of the referenced section. There is no automatic cross-referencing.
- **revtex4-2 with biblatex.** The combination of `revtex4-2` and `biblatex` is incompatible due to conflicts in `\addtocontents` patching between the two packages. Although not a `citemsp` issue, users should keep that in mind. Make sure to use the `natbib` backend with `revtex`.

9 Conventions

- **Section numbering:** use the source's native format (e.g., 3.2.1, IV, A.1).
- **Paragraph counting:** count paragraphs from the start of the referenced section. This is a manual convention; `citemsp` does not verify paragraph numbers against the cited source.
- **Subsection references:** use the full path (e.g., `key/3.2.1`), not a relative number.
- **Plain citations:** `\citemsp{key}` with no slashes produces a standard numeric citation, so the command can serve as a drop-in replacement for `\cite` if desired.

10 Requirements

- `LATEX2e` (2020/10/01 or later)
- `biblatex` (any numeric style) **or** `natbib` (with the `numbers` option)
- `graphicx`
- `etoolbox`
- `amssymb` — recommended for best glyph quality ($\triangleq$, $\boxplus$). If not loaded, `citemsp` provides primitive fallback definitions.

11 License

This work is released under the [LaTeX Project Public License, version 1.3c](#) or later.

References

- [1] Ahimsa Campos-Arceiz. “Shit Happens (to be Useful)! Use of Elephant Dung as Habitat by Amphibians”. In: *Biotropica* 41.4 (2009), pp. 406–407. DOI: [10.1111/j.1744-7429.2009.00525.x](https://doi.org/10.1111/j.1744-7429.2009.00525.x).
- [2] David A. Cox and Steven Zucker. “Intersection Numbers of Sections of Elliptic Surfaces”. In: *Inventiones Mathematicae* 53 (1979), pp. 1–44. DOI: [10.1007/BF01403189](https://doi.org/10.1007/BF01403189).
- [3] Stefano Ghirlanda, Liselotte Jansson, and Magnus Enquist. “Chickens Prefer Beautiful Humans”. In: *Human Nature* 13.3 (2002), pp. 383–389. DOI: [10.1007/s12110-002-1021-6](https://doi.org/10.1007/s12110-002-1021-6).
- [4] Yuri Lazebnik. “Can a Biologist Fix a Radio? — Or, What I Learned While Studying Apoptosis”. In: *Cancer Cell* 2.3 (2002), pp. 179–182. DOI: [10.1016/S1535-6108\(02\)00133-2](https://doi.org/10.1016/S1535-6108(02)00133-2).
- [5] Robert A. J. Matthews. “Tumbling Toast, Murphy’s Law and the Fundamental Constants”. In: *European Journal of Physics* 16.4 (1995), pp. 172–176. DOI: [10.1088/0143-0807/16/4/005](https://doi.org/10.1088/0143-0807/16/4/005).
- [6] Hans C. Mayer and Rouslan Krechetnikov. “Walking with Coffee: Why Does It Spill?” In: *Physical Review E* 85 (2012), p. 046117. DOI: [10.1103/PhysRevE.85.046117](https://doi.org/10.1103/PhysRevE.85.046117).
- [7] Victor Benno Meyer-Rochow and József Gal. “Pressures Produced When Penguins Pooh — Calculations on Avian Defaecation”. In: *Polar Biology* 27 (2003), pp. 56–58. DOI: [10.1007/s00300-003-0563-3](https://doi.org/10.1007/s00300-003-0563-3).
- [8] Gordon Pennycook et al. “On the Reception and Detection of Pseudo-Profound Bullshit”. In: *Judgment and Decision Making* 10.6 (2015), pp. 549–563. DOI: [10.1017/S1930297500006999](https://doi.org/10.1017/S1930297500006999).
- [9] Gordon C. S. Smith and Jill P. Pell. “Parachute Use to Prevent Death and Major Trauma Related to Gravitational Challenge: Systematic Review of Randomised Controlled Trials”. In: *BMJ* 327 (2003), pp. 1459–1461. DOI: [10.1136/bmj.327.7429.1459](https://doi.org/10.1136/bmj.327.7429.1459).
- [10] Dennis Upper. “The Unsuccessful Self-Treatment of a Case of “Writer’s Block””. In: *Journal of Applied Behavior Analysis* 7.3 (1974), p. 497. DOI: [10.1901/jaba.1974.7-497a](https://doi.org/10.1901/jaba.1974.7-497a).
- [11] Liming Wang, Zdeněk Sofer, and Martin Pumera. “Will Any Crap We Put into Graphene Increase Its Electrocatalytic Effect?” In: *ACS Nano* 14.1 (2020), pp. 21–25. DOI: [10.1021/acsnano.9b00184](https://doi.org/10.1021/acsnano.9b00184).
- [12] Te Faye Yap et al. “Necrobotics: Biotic Materials as Ready-to-Use Actuators”. In: *Advanced Science* 9 (2022), p. 2201174. DOI: [10.1002/advs.202201174](https://doi.org/10.1002/advs.202201174).
- [13] Doug Zongker. “Chicken Chicken Chicken: Chicken Chicken”. In: *Annals of Improbable Research* 12.5 (2006), pp. 16–21. URL: <https://isotropic.org/papers/chicken.pdf>.